

Population Correlation Coefficient

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{(\sum (x_i - \bar{x})^2)(\sum (y_i - \bar{y})^2)}}$$

$\sigma_x, \sigma_y \rightarrow$ Population SD

$\sigma_{xy} \rightarrow$ Population Covariance

$\bar{x}, \bar{y} \rightarrow$ Population Mean

Sample Correlation Coefficient b/w X and Y

$$r_{xy} = \frac{s_{xy}}{s_x s_y} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{(\sum (x_i - \bar{x})^2)(\sum (y_i - \bar{y})^2)}}$$

$s_x, s_y \rightarrow$ Sample SD

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$\bar{x}, \bar{y} \rightarrow$ Sample Mean

① Sample Correlation Coefficient:

$$r_{x,y} = \frac{\text{Cov}(X,Y)}{\sigma_x \sigma_y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

Cov is Covariance: $\text{Cov}(X,Y) = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{N}$

② Population Correlation Coefficient:

$$\rho_{x,y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

$n \rightarrow$ sample size

~~x_i, y_i~~

$x_i, y_i \rightarrow$ i th sample points

$\bar{x}$ & $\bar{y} \rightarrow$ sample means for random variable X & Y

③ Linear Correlation Coefficient:

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

r is near 1 $\rightarrow$ 2 variables have strong linear relationship

r is near 0 $\rightarrow$ 2 variables have no linear relation.

r is near -1 $\rightarrow$ 2 variables have weak (-ve) linear relationship

We are using Pearson Correlation.

Determine linear relationship b/w 2 variables
Value lies b/w -1 and 1.

(8)

Q. Find Pearson Correlation Coefficient between x and y:

x	600	800	1000
y	1200	1000	2000

To simplify, divide both x and y by 100:

x/100	y/100	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
6	12	-2	-2	4	4	4
8	10	0	-4	0	16	0
10	20	2	6	4	36	12

$$\bar{x} = 8 \quad \bar{y} = 14$$

$$\text{Pearson Correlation Coefficient for population} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}} = \frac{16}{\sqrt{8} \sqrt{56}} = \frac{2}{\sqrt{7}} = 0.756$$

Q. Survey was conducted in your city. Given is following sample data containing a person's age and their corresponding income. Find out whether the increase in age has an effect on income using correlation coefficient formula.

Age	25	30	36	43
Income	30000	44000	52000	70000

To simplify, divide y by 1000:

Age	Income/1000	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
25	30	-8.5	-19	72.25	361	161.5
30	44	-3.5	-5	12.25	25	17.5
36	52	2.5	3	6.25	9	7.5
43	70	9.5	21	90.25	441	199.5

$$\bar{x} = 33.5 \quad \bar{y} = 49$$

Calculate the Correlation Coefficient of given data:

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X 41 42 43 44 45
y 3.2 3.3 3.4 3.5 3.6

$n=5$

X	y	xy	x ²	y ²
41	3.2	131.2	1681	10.24
42	3.3	138.6	1764	10.89
43	3.4	146.2	1849	11.56
44	3.5	154	1936	12.25
45	3.6	162	2025	12.96
$\Sigma x = 215$	$\Sigma y = 17$	732	9255	57.9

$$\Sigma (x - \bar{x})^2 = 10 \quad \Sigma y = 732$$

$$\Sigma (y - \bar{y})^2 = 0.1$$

$$r = \frac{\Sigma ((x - \bar{x})(y - \bar{y}))}{\sqrt{(\Sigma (x - \bar{x})^2)(\Sigma (y - \bar{y})^2)}} = \frac{1}{\sqrt{(10)(0.1)}} = 1$$

Significant relationship b/w x & y

Q. Suppose S is a finite or countable set. Is it possible that $P(\{s\}) = 0$ for every single $s \in S$? (6)

$\Rightarrow$ if S is finite:

Then by additivity, the probability of whole sample space is:

$$P(S) = \sum_{s \in S} P(\{s\})$$

If every single $P(\{s\}) = 0$ then sum is 0 which contradicts the fact $P(S) = 1$

So for finite $S \rightarrow$ it is impossible

$\Rightarrow$ If S is countably finite

$$P(S) = \sum_{s \in S} P(\{s\})$$

if each term is zero, whole sum is 0 But we need $P(S) = 1$

So for countably infinite $S \rightarrow$ also impossible

Q. Suppose S is uncountable set. Is it possible that $P(\{s\}) = 0$ for every single $s \in S$?

$\Rightarrow$

$S = [0, 1]$ with uniform distribution

1) Probability of any single point is $P(\{s\}) = 0$

2) For interval $[0.2, 0.5]$: 0.3

3) Probabilities are assigned not to points but to intervals

In an uncountable set: Probability is not defined by summing up singletons, but by integrating over sets, An uncountable sum of zero can still add up to 1

Yes, possible when S is uncountable

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$n \rightarrow$ sample size

~~x_i & y_i~~

x_i & $y_i \rightarrow$ i^{th} sample points

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